Week 01: Foundations of Machine Ethics and AI Safety

Machine Ethics and Artificial Intelligence Safety

Machine Ethics and Artificial Intelligence Safety (11118BLG001). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

The course opens with a question that has accompanied computing since its early decades: What happens when machines act on objectives that their designers only partly understand? This week traces the question from Wiener's warning about automation to present research programmes in machine ethics and AI safety, and it introduces the vocabulary used throughout the course [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to distinguish machine ethics from AI safety, AI security and AI alignment. They should be able to classify artificial agents with Moor's four-level typology. They should also explain why rule lists such as Asimov's laws fail as a design method, and describe the research agendas that structure the rest of the course.

Lecture

Two fields with a shared origin

In 1960, Norbert Wiener argued that machines able to learn could pursue their programmed purposes in ways their users would not have chosen. He added that the speed of such machines could leave too little time for human correction [1]. The same concern now appears under two research labels. Machine ethics asks how artificial agents can represent and apply moral considerations when they act [2, 4]. AI safety asks how learning systems can be kept from unintended and harmful behaviour, especially when their objectives, training data or operating conditions are imperfect [3, 5].

The two labels overlap but emphasise different questions. Machine ethics is closer to moral philosophy: It studies which values a system should respect and how moral reasoning can be implemented. AI safety is closer to engineering and statistics: It studies failure modes such as reward hacking, distribution shift and adversarial manipulation. Kose argued that the two

agendas belong together. Dystopic scenarios, moral dilemmas and technical failure modes all bear on one practical question, namely whether future intelligent systems will be safe enough [6]. Two further terms complete the vocabulary. AI security concerns deliberate attacks on AI systems by adversaries. AI alignment concerns the match between what a system pursues and what its principals intend [7, 8].

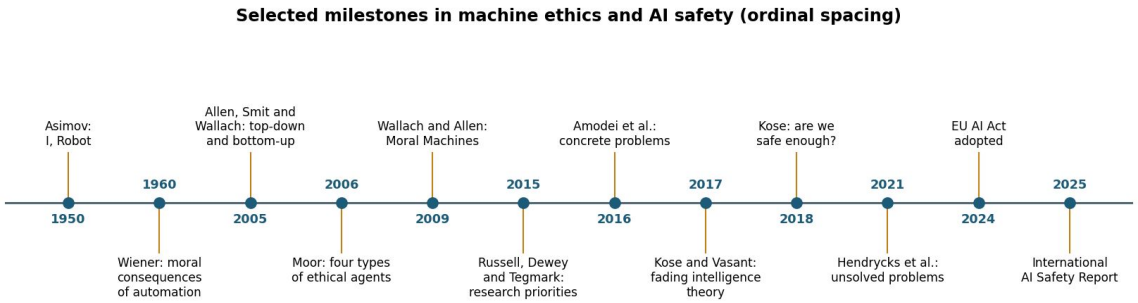


Figure 1.1. Selected milestones discussed this week, from Asimov and Wiener to the EU AI Act and the International AI Safety Report [1, 2, 3, 6, 9, 10, 11].

Moor's typology of ethical agents

Moor proposed four levels that remain useful for classifying systems [2]. An ethical impact agent affects people morally without representing ethics at all, as in his example of robot jockeys that replaced children in camel races. An implicit ethical agent has safety or fairness built into its design, for example an autopilot whose protections prevent dangerous manoeuvres. An explicit ethical agent represents ethical categories and reasons with them to select actions. A full ethical agent would exercise moral judgment with the understanding and responsibility attributed to adult humans. Most deployed systems are ethical impact agents or implicit ethical agents, whereas much of machine ethics research aims at explicit agents [4, 12].

The typology also clarifies responsibility. The moral behaviour of an implicit ethical agent is fixed by its designers, so accountability for failures stays with the people who specify its constraints. Explicit ethical agents raise harder questions, because their decisions follow from internal reasoning that designers do not fully script. Floridi and Sanders argued that artificial agents can be sources of moral action even when they are not morally responsible in the human sense, which separates accountability from blame [13].

> ***Pause and reflect.*** *Pick two systems used every day, such as a navigation application and a language model assistant. Where does each belong in Moor's typology, and which feature of the system decides the answer?*

Why rule lists are not enough

Asimov's Three Laws of Robotics are often cited as a first attempt to specify machine ethics. Asimov, however, used them as a narrative device whose loopholes drove the plots of his stories

[9]. The laws illustrate three recurring problems. First, natural-language terms such as harm are vague, so every application requires interpretation. Second, rules conflict, and a fixed priority order produces counterintuitive results in edge cases. Third, rules are silent about situations their authors did not anticipate. Yigit, Kose and Sengoz examined how such rules translate into robotics rights and ethics rules and found the same tension between clarity and coverage [14].

These problems are not confined to fiction. Allen, Smit and Wallach described top-down approaches, which derive behaviour from explicit principles, and bottom-up approaches, which learn behaviour from experience, and argued that neither is sufficient alone [15]. The same tension reappears in AI safety as the gap between a specified objective and the behaviour that designers actually wanted [3]. The interactive lab and the Colab notebook of this week make the conflict problem concrete. A small robot follows three prioritised rules. Changing the order of the rules changes its verdicts, in ways that are easy to predict for some cases and surprising for others.

A research map for the course

Amodei and colleagues grouped concrete technical problems into avoiding negative side effects, avoiding reward hacking, scalable oversight, safe exploration and robustness to distributional shift [3]. Hendrycks and colleagues later organised open problems into robustness, monitoring, alignment and systemic safety [5]. Russell, Dewey and Tegmark had earlier set out research priorities that combined short-term questions about law, economics and verification with long-term questions about control [16]. Kose offered an introduction to AI safety for a broad audience [17]. Kose and Vasant proposed a fading intelligence theory, in which the operational lifetime of a system is limited so that newer and safer generations can replace it [18].

The course follows this map. Weeks 2 to 4 cover normative foundations, implementations of machine ethics and the aggregation of moral preferences. Weeks 5 and 6 address fairness. Weeks 7 to 10 cover specification, robustness, uncertainty and interpretability. Weeks 11 to 13 examine the alignment of large language models, their security and advanced risks such as deceptive behaviour. Week 14 closes with governance, standards and auditing.

Hands-on components

Interactive lab: Part A asks for Moor's level of eight systems and explains each case after checking. Part B gives a service robot three prioritised rules. Changing the priority order shows how strongly each verdict depends on it [2, 9]. Link: https://utkukose.github.io/machine-ethics-ai-safety-NB-lecture/weeks/week-01/lab.html

Colab notebook: It implements the three-rule robot of the interactive lab, enumerates all six priority orders and compares a lexicographic agent with a weighted-sum agent. The comparison shows that a ranking of rules and a weighting of rules are different design commitments [15].

Link: https://colab.research.google.com/github/utkukose/machine-ethics-ai-safety-NB-lecture/blob/main/weeks/week-01/NB01_foundations.ipynb

Reflection prompts

1. Describe one AI system that you use or develop. Which level of Moor's typology does it occupy, and who is accountable when it fails?
2. Which problem of rule lists (vagueness, conflict or unanticipated situations) seems most dangerous in your research area, and why?
3. Write down one question about machine ethics or AI safety that you expect this course to answer. Keep it and revisit it in Week 14.

Weekly task and submission

What to do. Write a short position statement on the vocabulary of the course. First, define machine ethics, AI safety, AI security and AI alignment in your own words, one short paragraph each. Second, choose one system from your own field and place it in Moor's typology, justifying the level with one feature of the system. Third, name one risk of the system and state which of the four fields would study it.

Where to look. Lecture: sections "Two fields with a shared origin" and "Moor's typology of ethical agents". Interactive lab: Part A, where eight systems are classified with Moor's levels. Colab notebook: Sections 1 and 2, which show why the order of rules matters.

Example. A hospital triage assistant that ranks patients by urgency is an implicit ethical agent: Its designers built safety rules into the ranking, but the system does not reason with ethical principles. Wrong rankings for rare conditions would be a question for AI safety.

What to submit. About 400 words that cite at least three works from this week's references in square brackets, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

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